

Preregistration

#Knowledge

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Study Information

Research Question	Is the seeding procedure useful as an easy-to-implement and large-scale intervention in an ecologically valid everyday context such as social media?
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Description	Training <i>carbon footprint</i> and <i>calorie</i> estimation in social media
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Hypotheses	1. Increase in estimation accuracy (metric knowledge) for matching seeding and estimation dimensions. 2. Increase in rank correlation (mapping knowledge) for matching seeding and estimation dimensions.
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Exploratory:

- 1) Greater seeding effect when participants saw more posts (as indicated by the number of liked posts)
- 2) Greater seeding effect for domain with less prior knowledge (carbon footprint) vs. domain with more prior knowledge (calorie content)

Experiment

Design The study will have a 2 (*seeding dimension*: carbon footprint vs. calorie content) by 2 (*estimation dimension*: carbon footprint vs. calorie content) by 2 (*target fact item type*: seeding vs. non-seeding) design

Materials Three different types of items: *target fact items (transfer and seeding items)* vs. *trivia fact items*.

1. **Target fact items:** 60 food items (the *same* for both seeding dimensions) with information about carbon footprint and calorie content (which is shown to the participants depends on seeding dimensions). Target fact items are selected based on the following criteria:
 - From one out of five possible categories of food (fruit & vegetables, meat & fish, dairy, grain products, sweets)
 - Common food items
 - Information about carbon footprint and calorie content available.
- 1.1. **Seeding items:** For each seeding dimensions, 15 seeding items will be selected from the 60 food items, based on the following criteria:
 - Three items out of each food category
 - Representative Mean, SD, and range of criterion as the complete item set
- 1.2. **Transfer items:** The remaining 45 target fact items not selected as seeding items (i.e., the transfer items) will be only shown in the final testing phase.

2. **trivia fact items:** 21 (15 during study + 6 for old-new recognition test) not carbon footprint or calorie related content about food (the *same* facts for each seeding set condition)

Procedure

The general procedure is outlined in the Figure below. The study will consist of *three* phases.

1. Recruiting Phase

In the recruiting phase of the study, we will collect E-Mail addresses of participants who are interested in participating in the study. Until enough people signed up for the study (see section *Sample Size*), already signed participants will receive weekly E-Mails telling them that they are still signed up for it and that the study will start soon. When enough people signed up, potential participants are invited to follow one of two Instagram accounts (Mosleh et al, 2022). Both accounts will be advertised as posting daily “general and fun science facts about food”. The content of the two accounts will be the same, except for the specific target facts seeding items shown on each account: Calorie content or carbon footprint of food items, intermixed with other trivia facts about food (identical in both conditions). Participants will be instructed to like each post they see. In addition, participants will also be informed that there will be a questionnaire to evaluate the contents of the account after some time and that the accounts will be deleted after this evaluation. After this instruction E-Mail, the seeding phase will start three days after.



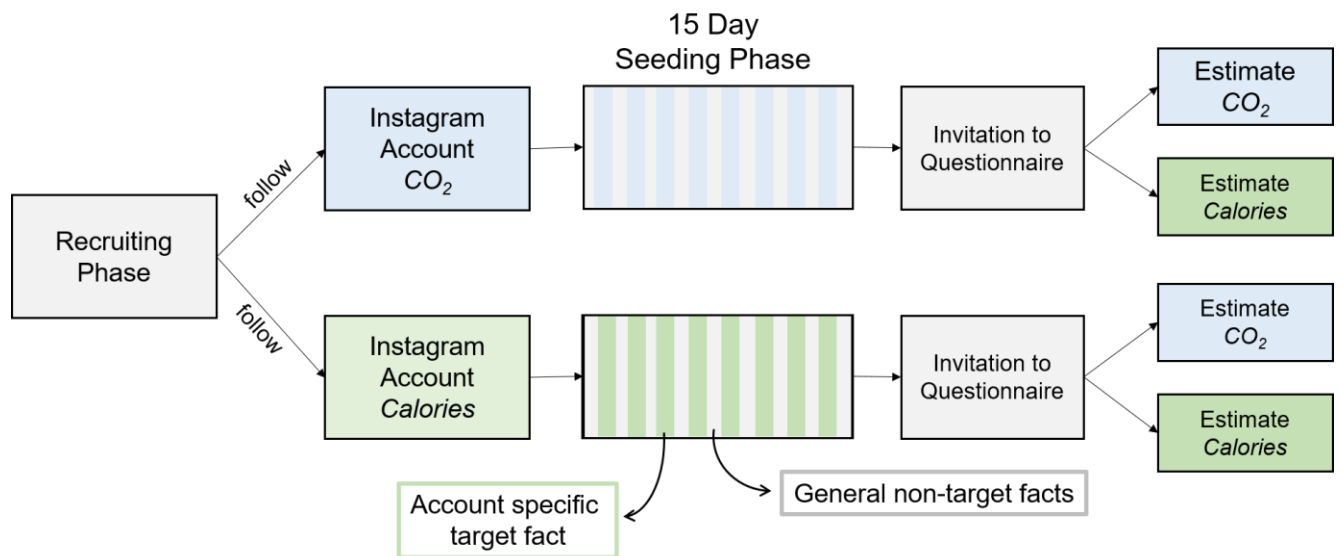
Example post of a target fact seeding item in the calorie content condition

2. Seeding Phase

During the 15 days seeding phase, we will post *two posts per day* on each account: one target post with a seeding item (see example) and one non-target post. The non-target post will be the same for each group. Posts will be published at 12:00 and 14:00, the order will be counter-balanced (i.e., target post first, non-target post second).

3. Testing Phase

After the 15 day seeding phase with 15 seeding items, participants will be invited to the final questionnaire. The invitations will be posted on the respective private Instagram accounts, in order to separate participants of the respective seeding set condition (carbon footprint vs. calorie content). In the questionnaire, participants will be asked to estimate either the carbon footprint or calorie content of all 60 food items (depending on the estimation set condition). This is, from the participants who follow the Instagram account showing the seeding facts about the carbon footprint, 50% will have to estimate the carbon footprint and the other 50% the calorie content and vice versa. After the estimation phase, participants will be presented with a two-alternative-forced choice task where they have to indicate for 6 pairs of old and new non-target facts, which one they have seen over the course of the 15 days. This will allow us to check whether participants actually have read and remember the posts. In addition, before the invitation to the final questionnaire, all previous posts on the accounts will be deleted in order to prevent participants from looking up the information posted on the accounts.



Variables

Manipulated variables

- **Seeding dimension:** carbon footprint vs. calorie content
 - **Estimation dimension:** carbon footprint vs. calorie content
-

Measured variables

Recruiting Phase

- Consent yes/no
- E-Mail addresses

Seeding Phase

- Date & Instagram handle of persons liking each post (updated daily)

Testing Phase

- Numerical estimation
 - Two-forced choice answers
 - Instagram handle in order to match data of post interaction to estimation data
 - Demographics (age, gender, knowledge about food and the respective areas, numerosity, education)
-

Indices (optional)

As dependent variables we will calculate Order of Magnitude Error (OME) and the rank correlation between estimated and actual criterion values. The OME is defined as:

$$|\log_{10}\left(\frac{Estiamted}{True}\right)|$$

Participants

Data collection procedures

Participants will be recruited through advertisements at the University of Mannheim, Social Media (e.g., via Instagram) and online forums (e.g., Reddit). Ten vouchers with 50€ each will be raffled between all participants who completed the final survey, want to participate in the raffle and are at least 18 years old.

Sample size

The seeding phase will be started when $N = 400$ participants signed up for the study or after three weeks of recruiting.

Sample size rationale

We determined the sample size for one group (i.e., one estimation dimension) with an *a priori* power analysis for a two-sample independent *t*-test (seeding dimension same as estimation dimension yes vs. no) assuming a Cohens $d = .50$ (conservative estimate based on Bröder et al., 2023; Wohldmann et al., 2015; Wohldmann & Healy, 2020), assuming $\alpha = .05$ and a Power of $1 - \beta = .90$. This results in a sample size of $n = 140$ for one estimation dimension, thus $N = 400$ is the sample size for both estimation dimension and additional participants which might drop out of the study. The hierarchical modeling approach we plan to use is more powerful than the two-sample independent *t*-test and thus our actual power with this sample size is probably a bit higher than assumed here.

Analysis Plan

The main analysis will be done only on the transfer items, this is, the seeding items, which only half of the participants actually saw for the respective estimation domain, will be excluded for the main analysis.

Statistical models

Estimation Accuracy: Give the rather skewed exponential distribution of absolute difference values and the possibility of data points equal 0, the data will be analyzed with two hierarchical log-normal hurdle model (one for each estimation domain) using the `brms` package in R. The following model syntax will be used:

```
AV ~ match_domain +  
      (match_domain|ID) +  
      (match_domain|item_ID)
```

```
Hurdle ~ match_domain
```

where,

- AV is the dependent variable (i.e., (log) absolute error between estimated and actual criterion values)
- `match_domain` specifies whether the seeding and estimation domain where the same, i.e. whether participants had the chance to learn about the criterion
- `(match_domain|ID)` is the random intercept and slope per person
- `(match_domain|item_ID)` is the random intercept and slope per specific food item

When there are no real 0 data points, we will omit the hurdle part of the model.

Priors: Based on prior studies, the following priors will be used

```
Intercept ~ Normal(-0.75, 0.50)  
bmatch_domain ~ Normal(-0.20, 0.15)  
σID/Person ~ InverseGamma(5,1)  
σe ~ InverseGamma(5,1)  
ρ ~ LKJ(2)
```

where ρ is the prior for the correlation between random slopes and intercepts.

Rank Correlation: To investigate the effect of seeding on the mapping knowledge as measured by the rank correlation, which can

range from -1 to 1, we will use a similar approach. However, instead of using a log-normal hurdle model we will use a standard gaussian model with the following priors:

$$\begin{aligned}\text{Intercept} &\sim \text{Normal}(0, 0.50) \\ b_{\text{match_domain}} &\sim \text{Normal}(0, 0.50) \\ \sigma_{\text{ID/Person}} &\sim \text{InverseGamma}(5,1) \\ \sigma_e &\sim \text{InverseGamma}(5,1) \\ \rho &\sim \text{LKJ}(2)\end{aligned}$$

Also, since the rank correlation is calculated over the trials of each person, the model syntax will be:

$$\text{AV} \sim \text{match_domain} + (\text{match_domain}|\text{ID})$$

Inference criteria

Hypothesis will be tested by computing the Bayes Factor between a model including the effect of `match_domain` (M_1) as fixed effect vs. a model without the fixed effect term (M_0). The models:

$$M_1: \text{AV} \sim \text{match_domain} + (\text{match_domain}|\text{ID}) + (\text{match_domain}|\text{item_ID})$$

$$M_0: \text{AV} \sim 1 + (\text{match_domain}|\text{ID}) + (\text{match_domain}|\text{item_ID})$$

Data exclusion

Participants will be excluded ...

- ... when their performance in the 2AFC task is not above 50% correctly recognized no old non-target fact
- ... when they did not like a single post
- ... if they do not finish the final study in a time limit of 30 minutes (the estimated finishing time is 5 to 20 minutes).

- ... who indicate non-compliance. That means they either indicate having cheated and or just having clicked through the experiment (Aust et al., 2013)
- ... who indicated that they used some aid (looking up answers, taking notes, asking other persons)
- ... who indicate technical problems or that the experiment did not work as intended.
- ... who make only two or less different estimates for all items in the estimation phase.
- ... who provided judgments in under 1000 ms for more than 10% of the items in the estimation phase, as such short estimation processes are implausible.

Trials will be excluded ...

- ... for which the judgment was provided in under 1000 ms will be excluded.

**Exploratory
analysis (optional)**

Exploratory analyses will be tested by including either the number of liked posts or the seeding domain as fixed effects in the model. Hypothesis will again be tested via the Bayes Factor comparing models with and without the corresponding fixed effects.